The Edge-Integrity of Cartesian Products

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Abstract

The edge-integrity of a graph measures the difficulty of breaking it into pieces through the removal of a set of edges, taking into account both the number of edges removed and the size of the largest surviving component. We develop some techniques for bounding, estimating, and computing the edge-integrity of products of graphs, paying particular attention to grid graphs.

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1 Introduction

The edge-integrity of a graph attempts to measure the disruption caused by the removal of edges from the graph. We define a *fracturing* of a graph G as any subgraph $G - S$ formed by the removal of some set S of edges from G. We denote by $m(G - S)$ the maximum order (number of vertices) of a component of $G - S$, and define $\varphi(S) := m(G - S) + |S|$. Barefoot, Entringer and Swart [1] defined the *edge-integrity* of graph G, denoted $I'(G)$, by

$$I'(G) := \min_{S \subseteq E(G)} \varphi(S).$$

A set S for which $\varphi(S)$ is minimized is called an *I'-set*, and the graph $G - S$ is called an *optimal fracturing*. A graph is said to be *honest* if its edge-integrity is equal to its order.

Barefoot, Entringer and Swart [1] proposed edge-integrity as a measure of how hard it is to disrupt a network thoroughly by edge failures. It is to be noted that, while the edge-integrity achieves a trade-off between the number of edges removed and the severity of the fracturing of the network, adding the number of edges removed to the number of vertices in the largest remaining component is somewhat arbitrary. Nevertheless this parameter is a natural analogue to vertex-integrity, where vertices rather than edges are removed, and it has proven to be interesting.

The following proposition gives the edge-integrities of several familiar graphs:

Theorem 1.1
1) For the complete graph K_n on n vertices, $I' = n$.
2) For the complete bipartite graph $K_{m,n}$, $I' = m + n$.
3) For the path P_n on n vertices, $I' = \lceil 2\sqrt{n} \rceil - 1$.
4) For the cycle C_n on n vertices, $I' = \lceil 2\sqrt{n} \rceil$.

The following proposition gives some properties of edge-integrity. We use $\Delta(G)$ to denote the maximum degree of a vertex of graph G, and $\delta(G)$ the minimum degree.

Theorem 1.2 *For any graph G on p vertices,*
1) $I'(G) \geq \Delta(G) + 1$;
2) If G is connected then $I'(G) \geq 2\sqrt{p} - 1$;
3) If G has diameter 2 then $I'(G) = p$.

Part 2 was proved by Barefoot et al. [1] and part 3 by Bagga et al. [2]. Edge-integrity was also investigated by Fellows and Stueckle [4] and Laskar, Stueckle and Piazza [6]. There is also a survey on edge-integrity [3].

In this paper we investigate the edge-integrity of cartesian products of graphs, beginning in Section 2 with general bounds in terms of the edge-integrity and other parameters of the constituent graphs. In Section 3 we compute the edge-integrity of the subdivisions of the complete and complete bipartite graphs. These results are then used in Section 4 to determine the asymptotics of the edge-integrity of the product of two stars.

In Section 5 we describe a technique, involving boundary inequalities, which can be used to prove good lower bounds on the edge-integrity of many graphs. This technique is applied in Section 6 to the product of two paths where one path is much longer than the other, and in Section 7 to two (or more) paths of roughly the same length. These sections also consider the edge-integrity of related products. We conclude with some thoughts on future directions one might pursue.

2 General Bounds

We begin by considering general bounds on the edge-integrity of the cartesian product of two graphs. Recall that for two graphs G and H, the cartesian product $G \times H$ has vertex set $V(G) \times V(H)$, and (u_1, u_2) and (v_1, v_2) are adjacent if either $u_1 = v_1$ and u_2v_2 is an edge of H, or $u_2 = v_2$ and u_1v_1 is an edge of G.

An upper bound was observed in [2]:

Theorem 2.1 *For any graphs G and H, $I'(G \times H) \leq I'(G) \cdot |V(H)|$.*

In [2] it was also shown that equality holds if H is complete:

Theorem 2.2 *For any graph G, $I'(G \times K_n) = nI'(G)$.*

In particular, the cubes are honest.

Note that $I'(G \times K_n) = I'(G) \cdot I'(K_n)$. But there are graphs where the edge-integrity of the product is less than the product of the individual edge-integrities, and some where it is greater. For example, $I'(P_4 \times P_4) = 12$ while $I'(P_4) = 3$. On the other hand, $I'(P_3 \times K(1,3)) = 11$ while $I'(P_3) \cdot I'(K(1,3)) = 12$, as shown in Figure 1.

The latter example is also one where $I'(G \times H) < (\Delta(G) + 1) \cdot (\Delta(H) + 1)$. Nevertheless, Theorem 2.2 may be extended to obtain the following lower bound:

Theorem 2.3 *For graphs G and H, $I'(G \times H) \geq I'(G) \cdot (\delta(H) + 1)$.*

Proof. Let $G_1, G_2, \ldots, G_n$ denote the copies of G in $G \times H$, and let S be an I'-set of $G \times H$. For $i = 1, 2, \ldots, n$, let $S_i := S \cap E(G_i)$ and $m_i := m(G_i - S_i)$.

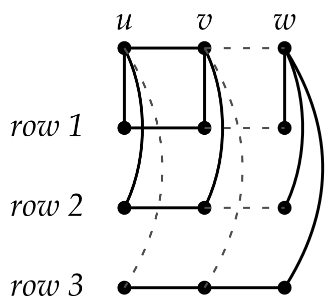

Figure 1: Optimal fracturing of $P_3 \times K(1,3)$

Assume that m_1 is the largest of the m_i, and let F be a largest component of $G_1 - S_1$. Assume that $G_2, G_3, \ldots, G_r$ are the copies of G that are adjacent to G_1. For $i = 2, 3, \ldots, r$ let e_i denote the number of edges between F and G_i which are in S.

Then $m(G \times H - S) \geq m_1 + \sum_{i=2}^{r}(m_1 - e_i)$. And $|S| \geq \sum_{i=2}^{r} e_i + \sum_{i=1}^{n} |S_i|$. Hence

$$\begin{aligned} I'(G \times H) &\geq m_1 + \sum_{i=2}^{r}(m_1 - e_i) + \sum_{i=2}^{r} e_i + \sum_{i=1}^{n} |S_i| \\ &\geq \sum_{i=1}^{r}(m_1 + |S_i|) \\ &\geq rI'(G). \end{aligned}$$

Since $r \geq \delta(H) + 1$, the theorem is proved. QED

Equality in the theorem is (only) attained if H is complete, or if H is the disjoint union of equal-sized cliques and $I'(G) = m(G)$.

3 Subdivisions

In this section we consider subdivisions of graphs in which each edge has been doubled to a path of length 2. The original reason for studying such graphs was that such subdivisions of complete bipartite graphs arise in the investigation of the edge-integrity of products of stars, but we have since found them to be of interest in themselves.

Given a graph G, let SG denote the *splice graph* of G which results from subdividing each edge of G exactly once. We call the original vertices *base* vertices and the new ones *splice* vertices. If G has p vertices and q edges, then clearly SG has p base vertices, q splice vertices, and $2q$ edges.

Consider an optimal fracturing of SG. If the two neighbors of a splice vertex w are in the same component of the fracturing, then w must also be in that component (since it adds at most 1 to the maximum component order as opposed to adding 2 to the edge-cut otherwise). Also, by similar reasoning we may assume that if the two neighbors of w are in different components, then w is in one of those components. We call a fracturing *grouped* if it has no isolated splice vertices. We have established the following result:

Lemma 3.1 *For graph G there is an optimal fracturing of SG which is grouped.*

We first consider the splice graph SK_p of the complete graph. The following notation will be useful. Let $\pi : [p_1, p_2, \ldots, p_k]$ denote a (integer) partition of the integer p, and for any $J \subseteq \{1, 2, \ldots, k\}$ let $P_J := \sum_{i \in J} p_i$. Define

$$M(\pi) := \max_{J \neq \emptyset} \left\lceil \frac{P_J(P_J + 1)}{2|J|} \right\rceil .$$

Note that if a collection of components in a grouped fracturing has P base vertices, then the collection must have at least $P(P+1)/2$ vertices. Hence, if π denotes the partition of the base vertices, then there must be a component of order at least $M(\pi)$. The following lemma shows that maximum component order $M(\pi)$ is always achievable:

Lemma 3.2 *Let π be a partition of p. Then there exists a grouped fracturing of SK_p in which the ith component has p_i base vertices and the maximum component order is $M(\pi)$.*

Proof. Partition the base vertices of SK_p into sets V_i of cardinality p_i. Start with S_i being the set containing V_i and those splice vertices which join pairs of vertices in V_i. We will distribute the remaining splice vertices into these S_i in such a way that each splice vertex has one of its neighbors in the same set, and no set has more than $M = M(\pi)$ vertices in all.

Assume that splice vertex w has not been distributed. Say the two sets containing the neighbors of w are S_1 and S_2. If one of these sets contains fewer then M vertices, then we can add w to it without problems. So assume that $|S_1| = |S_2| = M$.

By the definition of M, there must exist a splice vertex w_2 in $S_1 \cup S_2$ that is adjacent to a vertex of V_i for some $i > 2$. Say $i = 3$. If $|S_3| = M$ then there must similarly exist a splice vertex w_3 in $S_1 \cup S_2 \cup S_3$ such that w_3 is adjacent to a vertex of V_j for $j > 3$. Eventually we must reach an S_l for which $|S_l| < M$. In so doing we construct a sequence of splice vertices w_i $(2 \leq i \leq l-1)$ such that $w_i \in S_1 \cup S_2 \cup \cdots \cup S_i$ and w_i is adjacent to a vertex of V_{i+1}. Then we can shift w_{l-1} to S_l from say S_j, then shift w_{j-1} to S_j, and so on, eventually making room for w

p	I'	partition
2	3	any
3	5	several
4	9	several
5	13	several
6	18	$[3,1,1,1]$

p	I'	partition
7	24	$[3,1,1,1,1]$
8	31	$[3,1,1,1,1,1]$
9	39	$[5,2,2]$
10	48	$[5,2,2,1]$
11	58	$[5,2,2,1,1]$

Table 1: $I'(SK_p)$ for small p

in either S_1 or S_2. Hence we can finish distributing the splice vertices as required. QED

Theorem 3.3 *With notation as above,*

$$I'(SK_p) = p^2/2 - \max_{\pi} \left(\sum p_i^2/2 - M(\pi) \right).$$

PROOF. Let S be the set of edges removed to form an optimal fracturing of SK_p. We may assume that this fracturing is grouped. Let p_i denote the number of base vertices in the ith component. Then the size of S is $(\sum p_i(p-p_i))/2 = (p^2 - \sum p_i^2)/2$. Also, the maximum component order is at least $M(\pi)$. On the other hand, by the above lemma, any particular π gives rise to a grouped fracturing with p_i base vertices in the ith component and maximum component order $M(\pi)$. QED

Unfortunately, it is not in general a simple matter to compute the expression for $I'(SK_p)$ given in this theorem. It can, of course, be done by brute force for small values of p. Computation is aided by observing that if the partition $\pi = [p_i]$ of p is given in nonincreasing order, then to calculate $M(\pi)$ we need only consider J of the form $\{1, 2, \ldots, h\}$. For $1 \leq h \leq k$ we may define $P_h := \sum_{i \leq h} p_i$ and then

$$M(\pi) = \max_h \left\lceil \frac{P_h(P_h+1)}{2h} \right\rceil.$$

Table 1 lists the first few values and the partitions that achieve them.

We turn now to finding the asymptotic value. We need to estimate the maximum of $\sum p_i^2 - 2M(\pi)$ subject to the above conditions. We modify this in three ways:
(i) remove all integer restrictions on p_i;
(ii) redefine M as $\max_h P_h^2/(2h)$; and
(iii) normalize by setting $x_i := p_i/p$.
(Note that steps (i) and (ii) can only increase the maximum.)

This leaves us with the maximization problem in the following lemma:

Lemma 3.4 *Assume* $x_1 \geq x_2 \geq \ldots \geq x_k \geq 0$ *with* $\sum_{i=1}^k x_i = 1$*, and let* $S_h := \sum_{i=1}^h x_i$ *and* $A(\underline{x}) := \max_h S_h^2/h$*. Then*

$$\max_{k,x_i} \left(\sum_{i=1}^k x_i^2 - A(\underline{x}) \right) = 2(4 - \sqrt{2} - \sqrt{6})/3 =: \alpha,$$

which occurs for $k = 3$ *and* $x_i = (\sqrt{i} - \sqrt{i-1})/\sqrt{3}$ *(*$1 \leq i \leq 3$*) (and* $A = 1/3$*).*

Proof. Suppose that we have a feasible solution to this problem for a fixed value of k. Then, by convexity, we may assume that $S_h^2 = \min(1, hA)$ for all h. Consequently, in searching for the overall maximum we may assume that $1/(k-1) \leq A \leq 1/k$, and that $x_1 = \sqrt{A}$, $x_2 = \sqrt{2A} - \sqrt{A}$, $\cdots$, $x_{k-1} = \sqrt{(k-1)A} - \sqrt{(k-2)A}$ and $x_k = 1 - \sqrt{(k-1)A}$.

Letting $F_k(A) := \sum_{i=1}^k x_i^2 - A$, we therefore have

$$F_k(A) = A \left(\sum_{i=2}^{k-1} \left(\sqrt{i} - \sqrt{i-1} \right)^2 \right) + \left(1 - \sqrt{(k-1)A} \right)^2.$$

Since $F_k(A)$ is a quadratic in $\sqrt{A}$, it follows that its maximum occurs at one of the two extreme values of A. From this we may deduce that the overall solution equals the maximum of $\frac{1}{k} \sum_{i=2}^k \left(\sqrt{i} - \sqrt{i-1} \right)^2$ over k. This maximum occurs at $k = 3$ and has the value stated. QED

The next theorem shows that the edge-integrity of SK_p is about 91% of its order:

Theorem 3.5 *Let* $\gamma := (2(\sqrt{2} + \sqrt{6}) - 5)/3$ *(*≈ 0.909*). Then* $I'(SK_p) = \gamma\, p^2/2 + O(p)$*.*

Proof. From the definition of edge-integrity and the discussion preceding the above lemma, we have that $I'(SK_p) = p^2/2 - \max_\pi (\sum p_i^2/2 - M(\pi)) \geq p^2/2 - \alpha\, p^2/2 = \gamma\, p^2/2$. On the other hand, consider the partition π formed by taking $k = 3$ with $p_1 = \lfloor p/\sqrt{3} \rfloor$, $p_2 = \lfloor \sqrt{2}p/\sqrt{3} \rfloor - \lfloor p/\sqrt{3} \rfloor$ and $p_3 = p - \lfloor \sqrt{2}p/\sqrt{3} \rfloor$. Then $M(\pi) = p^2/6 + O(p)$ and $\sum p_i^2 = p^2/3 + (\sqrt{2}-1)^2 p^2/3 + (\sqrt{3} - \sqrt{2})^2 p^2/3 + O(p)$. Thus $I'(SK_p) \leq \gamma\, p^2/2 + O(p)$. QED

We turn now to the subdivision $SK_{r,s}$ of the complete bipartite graph. The analysis here is similar to the above, although computing exact values is a bit more complicated since it involves partitions of two integers. Because of the similarity, we omit many details here.

Let the partite sets of $K_{r,s}$ be A and B having orders r and s respectively. Let π be a double partition of the pair (r, s) given by $[(r_1, s_1), (r_2, s_2), \ldots, (r_k, s_k)]$ with

$r_i \geq 0$, $s_i \geq 0$, $\sum_{i=1}^k r_i = r$, $\sum_{i=1}^k s_i = s$, and if $r_i = 0$ then $s_i = 1$ and vice versa. For a subset J of $\{1, 2, \ldots, k\}$ let $R(J) := \sum_{j \in J} r_j$ and $S(J) := \sum_{j \in J} s_j$. Finally let

$$M(\pi) := \max_{J \neq \emptyset} \left\lceil \frac{R(J)S(J) + R(J) + S(J)}{|J|} \right\rceil .$$

By a proof similar to that of Lemma 3.2 we can establish the following lemma:

Lemma 3.6 *For any double partition π of (r, s) as described, there is a grouped fracturing of $SK_{r,s}$ in which the ith component has r_i base vertices from A and s_i from B and the maximum component order is $M(\pi)$. This is the smallest the maximum component order can be, given the distribution of the base vertices.*

Theorem 3.7 $I'(SK_{r,s}) = rs - \max_{\pi} \left(\sum_{i=1}^k r_i s_i - M(\pi) \right)$.

Proof. Let π be a double partition of (r, s) and let F be a grouped fracturing of $SK_{r,s}$ for π achieving $M(\pi)$ as in the above lemma. The ith component has $r_i(s - s_i) + s_i(r - r_i)$ edges to other components; so the cut corresponding to F has $\frac{1}{2} \sum (r_i(s - s_i) + s_i(r - r_i))$ or $rs - \sum r_i s_i$ edges. So, $I'(SK_{r,s}) = \min_\pi (rs - \sum_{i=1}^k r_i s_i + M(\pi))$. QED

We note that there is a simple formula when one of the partite sets is small enough.

Corollary 3.8 *For $r \leq 4$, $I'(SK_{r,s}) = rs + 2$.*

For large graphs we have only asymptotic values as before:

Theorem 3.9 *Let $\gamma := (2(\sqrt{2} + \sqrt{6}) - 5)/3$ (≈ 0.909). Then $I'(SK_{r,s}) = \gamma rs + O(r + s)$.*

Proof outline: As we did for SK_p, we take the formula from Theorem 3.7, remove the integer restriction, ignore the "linear" terms, and normalize the variables, thereby modifying our problem to the following: Maximize $\sum_{i=1}^k x_i y_i - B$ where $x_i \geq 0$, $y_i \geq 0$, $\sum_{i=1}^k x_i = \sum_{i=1}^k y_i = 1$, and $B = \max_J (\sum_{i \in J} x_i)(\sum_{i \in J} y_i)/|J|$.

It is readily argued that for the maximum to occur one may take $x_i = y_i$; whence this problem reduces to the optimization problem solved in Lemma 3.4. The result therefore follows as before. QED

4 Stars and Stars

In this section we consider the problem of computing the edge-integrity of the cartesian product of two stars. The above results on subdivisions carry over to products of stars as follows. Let T_n denote the star $K_{1,n}$. Then $T_r \times T_s$ is isomorphic to the graph obtained from $SK_{r,s}$ by adding a new vertex that is adjacent to all its base vertices. We therefore obtain the following result as a consequence of Theorem 3.9:

Theorem 4.1 *Let* $\gamma := (2\sqrt{2} + 2\sqrt{6} - 5)/3$ *(*≈ 0.909*). Then* $I'(T_r \times T_s) = \gamma\, rs + O(r+s)$.

One may calculate exact values in the simple case $r = 2$: the next theorem shows that the edge-integrity of the graph $T_2 \times T_s$ is approximately 11/12 of its order.

Theorem 4.2 *For all* s, $I'(T_2 \times T_s) = \lceil 11(s+1)/4 \rceil$.

PROOF. Let $G = T_2 \times T_s$, and call each of the copies of T_2 except the central copy a *row.* Let the central T_2 be the path uvw and let the rows be $u_i v_i w_i$ correspondingly ($i = 1, 2, \ldots, s$).

To prove the upper bound, let $t := \lceil (3s-1)/4 \rceil$ and let S be the set of all edges uu_i and vv_i for $t+1 \le i \le s$, together with the edge vw, and the edges $v_i w_i$ for $1 \le i \le t$. (See Figure 1 for the case $s = 3$ and $t = 2$.) Then $|S| = 2s - t + 1$. Also, the two components of $G - S$ have orders $2t+2$ and $3s - 2t + 1$, so that $m(G-S) = 2t + 2$ by our choice of t. Consequently $I'(G) \le \varphi(S) = 2s + t + 3 = \lceil 11(s+1)/4 \rceil$.

We prove that this value is a lower bound for $I'(G)$ by induction on s. That $T_2 \times T_1$ has edge-integrity 6, as required, is a consequence of Theorem 2.2. So assume the lower bound is true for $s - 1$ and consider s. Let S be an I'-set of G and let $F := G - S$. We may assume that F has no isolated vertex. Also, of the five edges incident with the vertices in a given row, we may assume that at most two are in S; for otherwise the row is isolated in F and then $3 + I'(T_2 \times T_{s-1}) = 3 + \lceil 11s/4 \rceil \ge \lceil 11(s+1)/4 \rceil$ is a lower bound on $\varphi(S)$. Hence we may assume that every vertex lies in a component with one of u, v or w, and thus F has either two or three components.

If there are three components, then each row has at least two of its five incident edges in S, so $|S| \ge 2s + 2$. On the other hand, the largest component of F has at least $s + 1$ vertices. So $\varphi(S) \ge 3s + 3$, which is too large.

Hence the fracturing F has just two components. Let H denote the one with two of the three vertices u, v and w, and let $h := |H|$. Since no row can have all five of its edges in F, each row has either three or four. Let t denote the number of rows with four edges in F. Each such row must have at least two of its vertices in H, so $h \geq 2t+2$. Furthermore, $|S| \geq 2s-t+1$ and $m(F) = \max(h, 3(s+1)-h)$. Therefore

$$\begin{aligned}
\varphi(F) &\geq 2s-t+1+\max(h,3(s+1)-h) \\
&\geq 2s-(h/2-1)+1+\max(h,3(s+1)-h) \\
&= 2(s+1)+\max(h,6(s+1)-3h)/2 \\
&\geq 2(s+1)+3(s+1)/4,
\end{aligned}$$

from which the result follows. QED

5 Boundary and Lower Bounds

In this section we show how to use isoperimetric inequalities and the like to establish lower bounds on the edge-integrity of a graph.

Suppose H is a fixed locally finite graph. If G is an induced subgraph of H, then the *boundary of G in H*, denoted $b(G)$, is defined to be the number of edges of H with exactly one end in G. Further, for n a nonnegative integer (not exceeding the order of H) we define $b(n)$ by

$$b(n) := \min\{\, b(F) : F \text{ is an induced subgraph of } H \text{ of order } n\,\},$$

(with $b(0) = 0$). We observe that if H is regular of degree d then $b(n)$ equals nd less twice the number of edges in a densest subgraph of H having order n.

Now, we define $B(m,p)$ to be the solution to the optimization problem:

> minimize $\sum_{i=1}^{k} b(p_i)$ over all k and all partitions $[p_1, p_2, \ldots, p_k]$ of p into k positive integers p_i with $\max p_i = m$.

Theorem 5.1 *Let G be a graph of order p and let H be a locally finite graph, with boundary function $b(n)$, containing G as an induced subgraph. Then*

$$I'(G) \geq \min_{m} \{m + B(m,p)/2\} - b(G)/2.$$

PROOF. Consider any set S of edges of G such that each of its edges joins vertices in different components of $G-S$. (Clearly any I'-set has this property.) Suppose the components of $G-S$ are $F_1, F_2, \ldots, F_k$ with orders $p_1, p_2, \ldots p_k$. Then $\sum_{i=1}^{k} b(F_i) = 2|S| + b(G)$. The result now follows from the definitions of $I'(G)$ and $B(m,p)$. QED

If the boundary function can be bounded from below by a function that is convex, we obtain the following corollary:

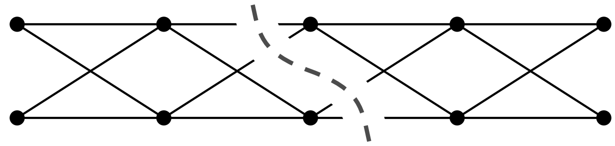

Figure 2: $P_{5(2)}$ is not honest

Corollary 5.2 *Let G be a graph of order p and let H be a locally finite graph, with boundary function $b(n)$, containing G as an induced subgraph. If there exists a real convex function $f(x)$ such that $f(n) \le b(n)$ for all $n \in \{1, 2, \ldots, p\}$, then*

$$I'(G) \ge \min_{x \ge 0} \left\{ x + \frac{p}{2x} f(x) \right\} - b(G)/2.$$

For more precise estimates, the following is sometimes useful:

Corollary 5.3 *Let G be a graph of order p and let H be a locally finite graph containing G as an induced subgraph. If the boundary function $b(n)$ of H is convex, then*

$$I'(G) \ge \min_{m} \{m + (q\, b(m) + b(t))/2\} - b(G)/2,$$

where q and t are defined by $p = qm + t$ with $0 \le t < m$.

As a first application, consider the expansion $P_r[\bar{K}_s]$ of the path defined by taking r disjoint sets $V_1, V_2, \ldots, V_r$ of s vertices each, and adding all edges between V_i and V_{i+1} for $i = 1, 2, \ldots, r-1$. We will also denote this graph by $P_{r(s)}$. Note that $P_{2(s)}$ is just the complete bipartite graph $K(s, s)$. The graph $P_{5(2)}$ is shown in Figure 2. The definition of $P_{\infty(s)}$ is clear by extension.

We will need the following easy lemma:

Lemma 5.4 *The graph $H = P_{\infty(s)}$ has boundary function*

$$b(n) = \begin{cases} 2sn - \lfloor n^2/2 \rfloor, & \text{for } n \le 2s, \\ 2s^2, & \text{for } n \ge 2s. \end{cases}$$

Proof. For F an n-vertex subgraph of H, let $R_1, R_2, \ldots, R_k$ denote the copies of $\bar{K}_s$ that F meets. Assume that R_i contains r_i vertices of F for $i = 1, 2, \ldots, k$. (Note $r_i \le s$.) If we take $r_0 = r_{k+1} = 0$ then it follows that R_i contributes $r_i(2s - (r_{i-1} + r_{i+1}))$ boundary edges to F. Hence by summing we find that $b(F) = 2sn - 2\sum_i r_i r_{i+1}$.

Let $n = sq + t$ with $0 < t \le s$. If $q > 1$ then $b(F)$ is a minimum when, for example, $k = q + 1$, $r_1 = r_2 = \cdots = r_q = s$ and $r_{q+1} = t$. In this case $b(F) = 2sn - 2(q-1)s^2 - 2st = 2s^2$. If $q \le 1$ then $b(F)$ is a minimum when, for example, $k = 2$, $r_1 = \lfloor n/2 \rfloor$ and $r_2 = \lceil n/2 \rceil$. QED

Theorem 5.5 *For all $r, s \geq 2$,*

$$I'(P_{r(s)}) \geq 2s\sqrt{rs} - s^2.$$

Equality holds if $r = k^2 s$ for some integer k.

Proof. Let $G = P_{r(s)}$ and $H = P_{\infty(s)}$. Note that $b(G) = 2s^2$. To prove the first part, by Theorem 5.1 we need to bound from below the expression

$$\min_m \{m + B(m, rs)/2\} - b(G)/2.$$

Suppose that $m \leq 2s$. Then, by convexity, $B(m, rs) \geq rs(2sm - m^2/2)/m = 2rs^2 - rsm/2$. For m in this range, the expression $m + rs^2 - rsm/4 - s^2$ is minimized at $m = 2s$. There its value is $s((r-2)(s-2)+2r)/2$, which is at least rs—the order of G.

Hence, by Corollary 5.3,

$$I'(G) \geq \min_{\substack{q \in \mathbb{N} \\ t \leq (rs-t)/q}} (rs - t)/q + qs^2 + b(t)/2 - s^2.$$

It can be verified that, for fixed q, this expression is minimized at either $t = 0$ or the maximum value of t (viz. $rs/(q+1)$), which amounts to t being 0 and q one more. So

$$I'(G) \geq \min_{q \in \mathbb{N}} rs/q + (q-1)s^2. \tag{1}$$

If we extend the range of q to the reals, then the minimum of this expression occurs at $q = s\sqrt{rs}$, and so $I'(G) \geq 2s\sqrt{rs} - s^2$, which establishes the inequality.

Now assume $r = k^2 s$. By making $k - 1$ cuts of s^2 edges each, one can split G into k blocks, each a copy of $P_{ks(s)}$. Then each component has order ks^2, so $I'(G) \geq (2k-1)s^2$. Since this is also the lower bound, equality holds. QED

Corollary 5.6 *$P_{r(s)}$ is honest if and only if $r \leq 2s$.*

Proof. If $r \leq 2s$ then the expression in Equation 1 above is minimized at $q = 1$, which proves that $P_{r(s)}$ is honest.

Now assume that $r \geq 2s + 1$. Then it is not hard to see that one can split G into two components of almost equal size by the removal of s^2 edges. The larger component will have order $\lceil rs/2 \rceil$. So $I'(G) \leq (rs+1)/2 + s^2 < rs$. QED

The analysis for the expansion $C_{r(s)}$ of the cycle, defined in the obvious way, is similar but simpler. Using $H = G$ we here find the following results, which we state without proof.

Theorem 5.7 *For $r \geq 3$ and $s \geq 2$,*

$$I'(C_{r(s)}) \geq 2s\sqrt{rs},$$

with equality if $r = k^2 s$ for some integer k. Also, $C_{r(s)}$ is honest if and only if $r \leq 4s$.

6 Stripes and Stripes

In this section we consider the value of $I'(G \times P_s)$ for G a fixed graph and s large. It is intuitive that a very good strategy is to remove "slices" between suitably chosen consecutive copies of G. We use the boundary approach to confirm this.

Lemma 6.1 *Let G be a λ-edge-connected graph on r vertices and let $H = G \times P_\infty$. Then for all n, $b(n) \geq \min(2r, \sqrt{2\lambda n})$.*

Proof. Let F be an induced subgraph of H of order n. Assume F meets x copies of G and y copies of P_∞. Then $b(F) \geq 2y$, and if $y < r$ then $b(F) \geq \lambda x$. Since $xy \geq n$, the bound follows. QED

Thus, if n is a sufficiently large multiple of r, $b(n) = 2r$.

Theorem 6.2 *Let G be a connected graph on r vertices. Then for s sufficiently large (say $s \geq 6r^2$),*

$$(2\sqrt{s} - 1)r \leq I'(G \times P_s) \leq (\lceil 2\sqrt{s} \rceil - 1)r = rI'(P_s).$$

Proof. The upper bound comes from Theorem 2.1. Let $H = G \times P_\infty$. As the lower bound for $b(n)$ in Lemma 6.1 is a convex function, it follows from Corollary 5.2 that the edge-integrity of $G \times P_s$ is at least the minimum of M_1 and M_2 where

$$M_1 := \min_{x \geq 0} \left\{ x + sr^2/x - r \right\}, \quad \text{and} \quad M_2 := \min_{x \geq 0} \left\{ x + rs/\sqrt{2x} - r \right\}.$$

Now, $M_1 \geq (2\sqrt{s} - 1)r$ and $M_2 \geq 3(rs)^{2/3}/2 - r$. For s sufficiently large $M_1 \leq M_2$, and the result follows. QED

Equality holds if s is a sufficiently large perfect square. But the answer is not, in general, the upper bound. For example, consider $G = P_4$, and s of the form $t^2 + 1$. Then a "straight slice" is one which removes 4 edges and separates two copies of G. A "skew slice" is one which removes 5 edges and breaks one copy of G into two parts that lie in separate components. If one uses $t - 2$ straight slices and 1 skew slice, then one can bring the maximum component order down to $4t + 2$. Thus $I'(G \times P_s) \leq 8t - 1$, whereas the upper bound of the above theorem is $8t$. See

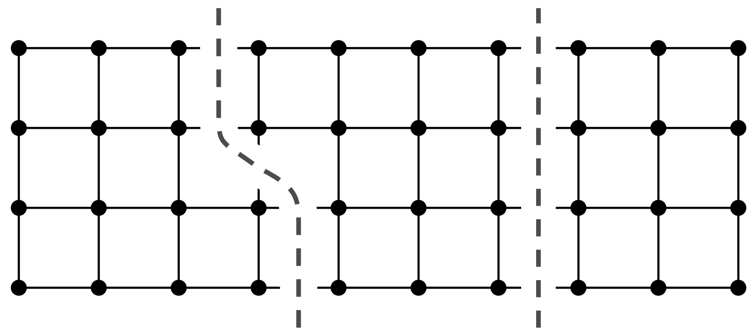

Figure 3: The I'-set of $P_4 \times P_{10}$ consists of one skew and one straight slice

Figure 3 for example. (This is in fact the correct value of the edge-integrity, as we shall see shortly.) A similar analysis can be performed for $G = P_r$ for $r = u^2$, and $s = t^2 + 1$ for $t \geq u$.

For some specific results, we start with G a path. Of course, $I'(P_2 \times P_s) = 2I'(P_s)$ by Theorem 2.2. We show next that the analagous result is true of $P_3 \times P_s$.

It is straight-forward to verify the following lemma:

Lemma 6.3 *The boundary function for the graph $P_3 \times P_\infty$ is given by*

$$b(n) = \begin{cases} n+2, & \text{if } 1 \leq n \leq 4, \\ 6, & \text{if } 3|n \text{ and } n \geq 6, \\ 7, & \text{otherwise.} \end{cases}$$

Some notation about partitions. For a partition $\pi = [p_1, p_2, \ldots, p_k]$ define

$$\mu(\pi) := \max_i p_i, \qquad \text{and} \qquad \sigma(\pi) := \textstyle\sum_i b(p_i).$$

Lemma 6.4 *Let $b(n)$ be the boundary function of $P_3 \times P_\infty$, and $\mu(\pi)$ and $\sigma(\pi)$ defined as above. Then the minimum of $\mu(\pi) + \sigma(\pi)/2$ over all partitions π of $3s$ is realized by a partition of the form $[3a, 3a, \ldots, 3a, 3b]$ for some $b \leq a$.*

Proof. Let π be an optimum partition of $3s$. Let $\pi = [p_1, p_2, \ldots, p_k]$ with $p_1 \geq p_2 \geq \ldots \geq p_k$. Let $M = \mu(\pi)$ and let $M = L + r$ where $0 \leq r \leq 2$ and L is a multiple of 3.

We show first that we may assume that at most one of the p_i is less than L. For otherwise proceed as follows: if $p_{k-1} + p_k \leq M$ then altering π by replacing the pair $\{p_{k-1}, p_k\}$ by their sum $p_{k-1} + p_k$ reduces the σ-value and therefore contradicts the optimality of π; and if $p_{k-1} + p_k > M$ then altering π by replacing the pair $\{p_{k-1}, p_k\}$ by the pair $\{p_{k-1} + p_k - L, L\}$ does not increase the σ-value and therefore yields an optimal partition with fewer values less than L. Hence we may assume that $p_{k-1} \geq L$.

We are done if $r = 0$ (note that p_k is then forced to be a multiple of 3). So suppose that $r \neq 0$. We will show that then we can always alter π so as either to decrease the value of $\mu(\pi) + \sigma(\pi)/2$ (a contradiction), or to change the μ-value to a multiple of 3. There are two cases:

1. $r = 1$: We first note that we may assume that $p_k \geq L$. For if not, we may increase p_k (by 1, 2, or 3) to the next higher multiple of 3 and reduce the corresponding number of $L+1$'s to L's, without increasing the σ-value (and repeat as necessary).

   Hence we may assume that π consists of a $L+1$'s and b L's where a is a positive multiple of 3. If $L = 3$, then replace $\{4, 4, 4\}$ by $\{6, 6\}$ to disprove the optimality of π. If $L \geq 6$ and $a = 3$, then replace $\{L+1, L+1, L+1\}$ by $\{L, L, L, 3\}$ to obtain an optimal partition of the desired form. If $L \geq 6$ and $a \geq 6$, then replace $\{L+1, L+1, L+1, L+1, L+1, L+1\}$ by $\{L+3, L+3, L, L, L, L\}$ to disprove the optimalty of π.

2. $r = 2$: We first note that we may assume that no $p_i \equiv 1 \pmod 3$. For suppose p_i is: then replacing the pair $\{p_1, p_i\}$ by $\{p_1 + 1, p_i - 1\}$ gives an optimal partition with μ a multiple of 3.

   Hence π has at least three elements $p_i \equiv 2 \pmod 3$ viz. p_1, p_2 and p_j (where $j = 3$ or k). Then replacing $\{p_1, p_2, p_j\}$ by $\{p_1 + 1, p_2 + 1, p_j - 2\}$ disproves the optimality of π. QED

Theorem 6.5 *For all s, $I'(P_3 \times P_s) = 3I'(P_s)$.*

Proof. By earlier results, we know the statement is true for $s \leq 3$; so we assume $s \geq 4$. By Theorem 5.1, the edge-integrity of $P_3 \times P_s$ is at least the minimum of $\mu(\pi) + \sigma(\pi)/2 - 3$ over all partitions π of $3s$. By the above lemma, the optimum π is given by $[3a, 3a, \ldots, 3a, 3b]$ for some $b \leq a$. If $a = 1$, then $\mu(\pi) + \sigma(\pi)/2 = 5s/2 + 3$. If $a \geq 2$, then $\mu(\pi) + \sigma(\pi)/2$ is

$$3a + 3\lceil s/a \rceil - \delta_{1b}/2.$$

Since the edge-integrity is an integer we may ignore the $\delta_{1b}/2$ term. Then the minimum of the expression over integral a is $3(\lceil 2\sqrt{s} \rceil - 1)$, and we are done. QED

A similar analysis for $P_4 \times P_\infty$, which we omit, yields the following results:

Lemma 6.6 *The boundary function for the graph $P_4 \times P_\infty$ is given by $b(n) = n + 2$ if $1 \leq n \leq 5$; $b(n) = n + 1$ if $6 \leq n \leq 7$; $b(n) = 8$ if $4|n$ and $n \geq 8$; and $b(n) = 9$ otherwise.*

Lemma 6.7 *Let $b(n)$ be the boundary function of $P_4 \times P_\infty$. Then the minimum of $\mu(\pi) + \sigma(\pi)/2$ over all partitions π of $4s$ is realized either by (i) a partition in which all the elements are multiples of 4, or (ii) one which is of the form $[4a+2, 4a+2, 4a, 4a, \ldots, 4a]$, or (iii) $[7, 7, 6]$.*

Theorem 6.8

$$I'(P_4 \times P_s) = \begin{cases} 4I'(P_s) - 1, & \text{if } s = t^2 + 1 \text{ or } s = t^2 + t + 1 \text{ for some } t \geq 2, \\ 4I'(P_s), & \text{otherwise.} \end{cases}$$

The same techniques work for $I'(C_r \times P_s)$. From Theorem 2.2 we know already that $I'(C_3 \times P_s) = 3I'(P_s)$, and the corresponding result holds for $C_4 \times P_s$ since $C_4 = K_2 \times K_2$. We omit the proofs of the following.

Lemma 6.9 *The boundary function for the graph $C_5 \times P_\infty$ is given by $b(n) = 2n+2$ if $1 \leq n \leq 3$; $b(n) = 8$ if $n = 4$; $b(n) = 10$ if $n = 6$ or $5|n$; and $b(n) = 12$ otherwise.*

Lemma 6.10 *Let $b(n)$ be the boundary function of $C_5 \times P_\infty$. Then the minimum of $\mu(\pi) + \sigma(\pi)/2$ over all partitions π of $5s$ is realized by a partition where all the elements are multiples of 5.*

Theorem 6.11 *For all s, $I'(C_5 \times P_s) = 5I'(P_s)$.*

The analogous result does not hold for $C_6 \times P_s$. (For example, $I'(C_6 \times P_7) < 42$.)

7 Grids

In this section we briefly consider general products of paths. Consider the k-dimensional grid-graph $G = P_{n_1} \times P_{n_2} \times \cdots \times P_{n_k}$. Let $N := \prod_{i=1}^{k} n_i$ and $S := \sum_{i=1}^{k} N/n_i$. The total number of edges in G is $kN - S$.

For a lower bound, we exploit the boundary approach using the supergraph $H = P_\infty^k = P_\infty \times P_\infty \times \cdots \times P_\infty$. We omit the easy proof of the following lemma:

Lemma 7.1 *The boundary function for the graph $H = P_\infty^k$ satisfies $b(n) \geq 2kn^{(k-1)/k}$.*

Theorem 7.2 *Let $G = P_{n_1} \times P_{n_2} \times \cdots \times P_{n_k}$, with $N = \prod_{i=1}^{k} n_i$ and $S = \sum_{i=1}^{k} N/n_i$. Then*

$$I'(G) \geq (k+1)N^{k/(k+1)} - S.$$

PROOF. Since the lower bound for the boundary function given in the above lemma is convex, and since $b(G) = 2S$, by Corollary 5.2, we have

$$I'(G) \geq \min_{x \geq 0} x + \frac{Nk}{x^{1/k}} - S.$$

By calculus, the minimum occurs at $x = N^{k/(k+1)}$, and the result follows. QED

An explicit constuction gives an upper bound of similar magnitude. Suppose for each i $c_i - 1$ "straight slices" are made in the ith dimension of G. Then the number of edges removed is $\sum_i N(c_i - 1)/n_i$. At the same time the maximum component size is reduced to $\prod_i \lceil n_i/c_i \rceil$.

If all the n_i are roughly the same then the best values of c_i are approximately $c_i = n_i/N^{1/(k+1)}$. In particular, for $P_n^k = P_n \times P_n \times \cdots \times P_n$ we obtain the following result:

Theorem 7.3 $I'(P_n^k) = (k+1)n^{k^2/(k+1)} - kn^{k-1} + O(n^{(k^2-k)/(k+1)})$.

For example, for $k = 2$ the theorem says that

$$I'(P_n \times P_n) = 3n^{4/3} - 2n + O(n^{2/3}).$$

Also, if the value of n is a $(k+1)$st power, we can determine the edge-integrity exactly:

Corollary 7.4 $I'(P^k_{a^{k+1}}) = (k+1)a^{k^2} - ka^{k^2-1}$.

8 Loose Ends

In the introduction we raised the question of whether vertices and edges should have the same importance in the definition of edge-integrity. Perhaps the terms $|S|$ and $m(G - S)$ should be assigned weights. While it is not clear (to us) what those relative weights should be, the boundary techniques can certainly be applied to the general α-edge-integrity defined by

$$I'_\alpha(G) := \min_{S \subseteq E}(|S| + \alpha \cdot m(G - S)).$$

This can probably be done in an even more general setting such as the schema of edge-integrity introduced in [5].

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